# CausalVerify: An Execution-Grounded Benchmark for LLM Causal Inference Workflows

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## Abstract

Existing causal-inference benchmarks for LLMs often evaluate text descriptions of methods or whether generated code can run. They rarely test whether the executed workflow recovers the target causal estimate. CausalVerify studies this verification problem by separating realistic interpretation from verifiable computation. The benchmark draws on 259 published economics papers, each represented by a reconstructed research question, data description, and institutional context, together with 100 fixed-seed synthetic scenarios that realise CSV datasets for difference-in-differences (DID), event study (ES), instrumental variables (IV), and regression discontinuity design (RDD). Experiment A scores method-family and direction agreement against four-LLM consensus labels. Experiment B runs model-written R code and checks whether the extracted treatment-effect estimate matches a canonical estimator on the same realised dataset. A calibration arm asks whether self-reported confidence separates correct from incorrect workflows. On Experiment B, seven evaluated LLMs reach final L2b+ pass rates between 10% and 88% at the default 50% tolerance (9% to 84% at a stricter 25%; Appendix C.4). Execution ranking (L2b) agrees with reference-agreement correctness (L2b+) far better than text-direction scoring (L4): Kendall $\tau = 0.81$ and Spearman $\rho = 0.93$, versus Kendall $\tau$ between $-0.20$ and $0.10$ for L4. A robustness run with Llama-3.3-70B-Instruct shows the same qualitative gap, and reported confidence does not reliably separate correct from incorrect workflows. Code, data, cached outputs, and a datasheet are released with the submission.

## 1 Introduction

Evaluating LLMs for empirical causal inference requires more than text scoring of methods and effect signs: it requires verifying that the executed analysis recovers a target estimate. A valid empirical claim requires an estimand, an identification argument, an estimator, executable analysis code, and a numerical result whose interpretation matches the target causal quantity [52, 50, 23, 49, 27, 4]. This structure exposes a failure mode that standard LLM evaluations can miss: a model may name the right empirical design, produce code that runs without error, and still estimate the wrong coefficient. In applied research, such failures are more dangerous than obvious failures because the response appears fluent, complete, and confident while the causal claim remains unverified. This concern sits within broader verification pressure in scientific publishing, where more than 10,000 papers were retracted in 2023, a record high reported by Van Noorden [61]. As LLMs move from text assistance toward software engineering, scientific-analysis agents, and laboratory-instrumentation control [17, 66, 55, 42, 62, 24, 30, 46, 15, 63], causal-inference evaluation must ask not only whether a model sounds right, but whether its executed workflow recovers the right estimate.

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Existing benchmarks capture only parts of this problem. Prior work and benchmarks on LLM causal reasoning study causal direction, graph reasoning, counterfactuals, correlation-to-causation judgments, intervention design, surveys of benchmark validity, and real-world identification tasks [34, 67, 32, 31, 14, 16, 65, 44, 68, 2, 38, 56, 54]. Code benchmarks evaluate whether programs can be synthesized, executed, or made to pass tests [13, 7, 22, 39, 43, 12, 36, 20, 30, 28, 69]. These settings are valuable, but causal workflows require a stricter check: after a model interprets a study, chooses a design, writes analysis code, and reports a treatment effect, does the executed workflow recover the intended estimate? This gap matters because code can pass surface checks while remaining semantically wrong [41].

We introduce CAUSALVERIFY, an execution-grounded benchmark for LLM causal-inference workflows. The benchmark deliberately separates ecological realism from executable verifiability. Real published papers provide realistic institutional context, but rarely provide the cleaned data, final runnable specification, and counterfactual treatment effect needed for direct verification. We therefore use 259 published economics papers to measure text-level agreement on method family and effect direction, not as verified causal correctness. Conversely, fixed-seed synthetic DGPs sacrifice some realism but provide executable targets: 100 realised CSV datasets across difference-in-differences, event study, instrumental variables, and regression discontinuity designs [25, 5, 59, 26, 37, 11, 45, 8, 1, 10, 19, 58, 53, 51]. Models write R code, the benchmark executes it, and L2b+ checks whether the extracted treatment-effect estimate matches a canonical estimator on the same realised dataset.

CAUSALVERIFY separates four signals that are often conflated: textual interpretation, code executability, numerical coefficient recovery, and self-assessed confidence. Across seven evaluated LLMs, final L2b+ pass rates range from 10% to 88%. Execution success is strongly aligned with reference-agreement correctness (Kendall $\tau = 0.81$, Spearman $\rho = 0.93$), whereas text-direction agreement is weakly or negatively aligned with Exp B L2b+ ranking (Kendall $\tau \in [-0.20, 0.10]$). Execution is not sufficient—some code runs and estimates the wrong coefficient—but it is a stronger proxy for execution-grounded agreement with the reference implementation under our reference-label construction than L4 direction agreement. A calibration arm further shows that naive retrospective confidence reports do not cleanly separate correct from incorrect workflows under our prompt, connecting the benchmark to work on calibration, verbalized uncertainty, selective prediction, and self-checking [21, 18, 29, 40, 48, 33, 35, 47, 60, 64]. Our contributions are a benchmark design separating real-paper agreement from execution-grounded reference-agreement correctness, L2b+ as a metric for agreement with a benchmark-fixed reference implementation, and empirical evidence that plausible text, runnable code, agreement with the reference implementation, and confidence can diverge sharply.

## 2 Related Work and Positioning

Prior work evaluates important pieces of the LLM causal-analysis workflow, but usually not the executed causal estimate. Causal benchmarks test textual reasoning about direction, counterfactuals, graphs, interventions, or identification. Code benchmarks test whether generated programs execute or pass functional tests. Science-agent benchmarks evaluate broader discovery or tool-use workflows. CAUSALVERIFY differs in its verification target: the primary correctness signal is not a text label or a generic program test, but whether executed model-written analysis code recovers a benchmark-defined target estimate on the realised dataset.

Among existing benchmarks, CauSciBench [2] is nearest in scope: it evaluates end-to-end causal-analysis performance over curated real, synthetic, and textbook tasks. CAUSALVERIFY asks a complementary measurement question: when a model appears to understand a causal task, which layer of the workflow is externally verified? It separates real-paper text agreement, controlled-DGP execution verification, and retrospective calibration rather than collapsing them into a single causal-ability score.

Concurrent work by Sawarni et al. [54] disentangles causal identification from estimation on real-world datasets, and its finding that strategy identification is easier than full specification is consistent with our name-versus-compute gap. The methodological difference is the unit of verification: CAUSALVERIFY verifies generated and executed R code on controlled DGP data, extracts the reported coefficient, and checks it against a realised-data canonical estimate. This isolates the failure

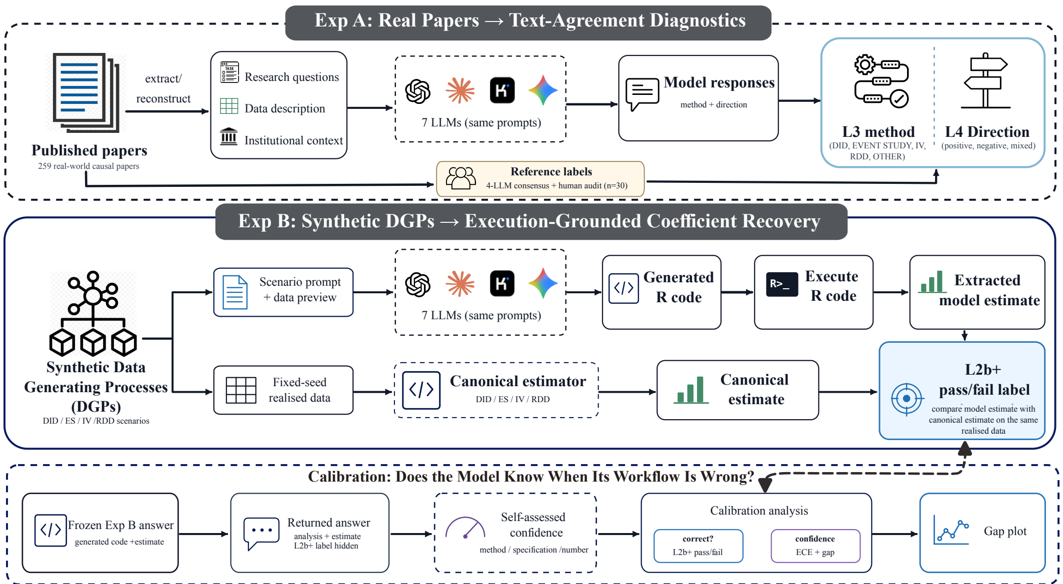


Figure 1: CAUSALVERIFY benchmark overview. Exp A uses published economics papers to form text-agreement tasks for method family and direction. Exp B uses fixed-seed synthetic DGPs and realised datasets to evaluate executable model-written R workflows against canonical estimators. The calibration arm returns each model its own Exp B answer and compares self-assessed confidence with L2b+ correctness.

mode where code runs but computes the wrong causal quantity. Recent workflow-oriented evaluations such as BioMysteryBench place models in bioinformatics environments with real data, tool access, and objective final answers [6]; CAUSALVERIFY is complementary: it fixes a realised-data causal reference estimate so that model-written causal code can be audited layer by layer rather than graded only on a final answer. Table 1 summarizes the distinction from adjacent benchmarks.

Table 1: Comparison with adjacent benchmarks. The table reports evaluation affordances rather than overall benchmark quality. A check mark (✓) indicates that a benchmark contains an explicit mechanism for that dimension, not that it is superior in all respects. ∼ = partial; ✗ = absent. †EconCausal counts 10,490 triplets across 2,595 studies (Lee et al. 2025).

| Benchmark | N | Target | Real ctx. | Exec. coeff. | Layered Dx | Self-assess. |
|---|---|---|---|---|---|---|
| CLadder / CORR2CAUSE [32, 31] | 11k / 400 | Causal QA / direction | ∼ | ✗ | ✗ | ✗ |
| CausalBench [68] | 3 | Graph + effect | ✗ | ✗ | ∼ | ✗ |
| EconCausal [38] | 10,490† | Economic causal sign | ✓ | ✗ | ✗ | ✗ |
| CauSciBench [2] | 367 | End-to-end CI pipeline | ✓ | ∼ | ∼ | ✗ |
| InterveneBench [56] | 744 | Intervention design | ✓ | ∼ | ∼ | ✗ |
| CausalReasoningBench [54] | 173 | ID + estimate | ✓ | ✗ | ∼ | ✗ |
| Code benchmarks [13, 30] | – | Program tests | ✗ | ∼ | ∼ | ✗ |
| Discovery / science agents [46, 15] | 264 / 102 | Discovery workflow | ✓ | ∼ | ∼ | ✗ |
| CAUSALVERIFY | 259 + 100 | Causal workflow | ✓ | ✓ | ✓ | ✓ |

## 3 Benchmark Construction and Controls

CAUSALVERIFY separates three evaluation regimes because no single setting provides both ecological realism and executable reference estimates. Real papers test whether models parse applied research contexts, but their reference labels are interpretive: the original cleaned data, exact runnable specification, and counterfactual effect are usually unavailable. This reflects the structure of empirical causal work, where claims depend on the alignment of estimand, identification, estimator, and data [4, 27]. Synthetic DGPs give up some realism but make coefficient correctness executable. Calibration then asks the deployment-facing question: when no external verifier is available, can the model's own confidence serve as a useful warning signal?

**Exp A: real-paper context.** Exp A contains 259 retained published economics papers after excluding two source failures. Each paper is represented by three reconstructed fields: research question, data description, and institutional context. Seven models are run on every paper, for 1813 outputs. Exp A is intentionally a text-level benchmark, comparable to prior LLM causal-reasoning evaluations that score textual causal direction, method recognition, sign judgments, or identification/estimation judgments [32, 31, 38, 54]. It measures method-family agreement (L3) and direction agreement (L4) against four-LLM consensus reference labels from a pool of Claude Opus 4.7, GPT-4o, Kimi, and Gemini 2.5 Flash; this pool overlaps with the evaluated panel and we discuss the implied structural circularity in ğ7. The benchmark does not treat these labels as human-adjudicated causal claims; it uses real papers to test whether models read applied study contexts in the same way as a structured consensus procedure. The execution-grounded reference-agreement endpoint is reserved for Exp B.

**Exp B: synthetic DGP execution.** Exp B contains 100 fixed-seed synthetic DGP scenarios across standard applied causal-inference designs: difference-in-differences and event studies, instrumental variables, and regression discontinuity designs [10, 58, 45, 25, 5, 59, 26, 37]. The 100 scenarios comprise 30 DID, 24 Event Study, 24 IV, and 22 RDD tasks; the realised datasets and canonical estimates are fixed before model evaluation. Models receive the research question, data description, and a preview of the realised CSV, then produce runnable R code. We execute the code and compare the extracted treatment-effect estimate to a canonical estimator computed on the same realised dataset. This realised-data comparison avoids penalizing models for finite-sample deviations between the realised sample and the ideal DGP parameter. Recent large-scale multi-analyst evidence shows that empirical claims can vary under alternative defensible analyses of the same research question and data [57, 9, 3]. We therefore treat the Exp B canonical estimator as a fixed reference path for executable evaluation, not as the only scientifically defensible analysis.

**Calibration: self-assessment.** We attempt a calibration query for each frozen Exp B output; response availability yields 646 calibration records. The model is shown its own previous analysis without the canonical estimate or pass/fail label and asked to report confidence in the method choice, specification, and numerical estimate. This is a retrospective verbalized-confidence protocol rather than token-probability calibration, connecting the benchmark to prior work on neural-network calibration, LLM self-knowledge, and confidence elicitation [21, 33, 60, 64]. The resulting confidence is compared to L2b+ correctness. This protocol deliberately mimics deployment: a user may see a model's confidence even when no canonical estimator is available.

**Controls and claim boundaries.** The benchmark uses a multi-stage evaluation pipeline rather than an autonomous causal scientist. Separate stages construct contexts, audit leakage, collect model responses, extract coefficients, aggregate labels, and elicit calibration, reducing "task writer = respondent = grader" circularity. The synthetic scenarios are not free-form LLM inventions: mathematical templates, treatment-effect signs, sample sizes, seeds, realised CSV files, and canonical estimators are fixed by the benchmark. Natural-language descriptions can vary, but a model cannot pass L2b+ by guessing the sign or recognizing a phrase; it must produce code whose estimated coefficient agrees with the realised-data reference. These components support different claims: Exp A is realistic but text-derived, Exp B is executable but synthetic, and calibration measures confidence–correctness alignment. The benchmark is strongest when these components are read together rather than collapsed into a single leaderboard.

## 4 Evaluation Layers

The benchmark uses six scoring labels, but they apply to different regimes. L1–L2b+ are Exp B workflow-implementation layers, while L3/L4 are Exp A text-agreement diagnostics. L1 checks that an output exists. L2a checks for a parseable R code block. L2b executes the code with `Rscript` in the fixed evaluation environment. L2b+ is the execution-grounded correctness layer: it checks whether the extracted treatment-effect estimate matches the realised-data canonical estimate within the default relative-error tolerance. L3 measures method-family agreement and L4 measures di-

Table 2: Benchmark controls. Each component supports a different claim and protects against a different failure mode.

| Component | Implementation | Claim supported |
|---|---|---|
| Real context | 259 papers; reconstructed RQ/DD/IC; consensus reference labels | Text-level method and direction agreement |
| Executable truth | 100 fixed-seed DGPs; realised CSVs; canonical estimators | Numerical coefficient correctness |
| Role separation | Separate stages for construction, response, labeling, extraction, calibration | Reduces circular grading |
| Leakage control | Method-name blacklist and prompt leakage audit | Prevents trivial label recovery |
| Scorer audit | Coefficient extraction judge plus human validation audit | Reduces brittle post-processing errors |
| Claim boundaries | Exp A = agreement; Exp B = execution; calibration = confidence alignment | Prevents overclaiming |

rection agreement against consensus reference labels. Only L2b+ is a numerical correctness layer; L1/L2a are bookkeeping diagnostics, L2b is execution, and L3/L4 are text agreement.

**Why L2b+ uses a canonical estimator.** The benchmark does not ask models to recover an ideal population parameter directly. In finite samples, even a textbook estimator may differ from the DGP parameter. We therefore compare model output to a canonical estimator applied to the same realised dataset. For scalar extracted estimate $\hat{\beta}_m$ and realised-data canonical estimate $\hat{\beta}_c$, L2b+ passes when

$$\frac{|\hat{\beta}_m - \hat{\beta}_c|}{|\hat{\beta}_c|} \leq \tau,$$

with default tolerance $\tau = 0.50$. Scenarios with $|\hat{\beta}_c| < 10^{-9}$ are marked unscoreable rather than divided. For Event Study scenarios, accepted post-event windows and scale conversions are frozen before scoring; valid cumulative post-event estimates are accepted only when they can be deterministically converted to the canonical per-day abnormal-return scale. This mapping is fixed before model comparison. Appendix C.1 lists the canonical reference estimate and accepted reported coefficient for each Exp B method family. L2b+ is therefore best read as agreement with a benchmark-fixed reference implementation, not as a verdict on causal-econometric correctness in general; a model that selects a methodologically more appropriate estimator (e.g. Callaway–Sant'Anna [10] or Sun–Abraham [58]) may diverge from the canonical reference, and we treat such divergence as benchmark coverage limitation rather than model error. All 30 Exp B DID scenarios use common adoption timing, in which Callaway–Sant'Anna and Sun–Abraham reduce to TWFE; Sun–Abraham implementations on three sampled scenarios pass L2b+ at 50% tolerance with relative errors below 32% (`audit/rebuttal_kit/cs_sa_estimator_check.csv`), confirming that the TWFE canonical reference is appropriate for this design space.

**Why L3/L4 are supporting metrics.** Exp A institutional context necessarily describes identifying variation, so method family can often be inferred by a trained reader. We therefore treat L3/L4 as agreement diagnostics, not verified causal correctness. They are useful diagnostics of how models parse real research contexts, but the load-bearing correctness evidence comes from Exp B.

**Scorer validation.** A preliminary 30-case audit of L2b-pass/L2b+-fail examples found that regex extraction produced many scorer-side failures. The final scorer therefore uses a separate coefficient-extraction judge to identify the reported treatment-effect estimate from executed stdout, followed by deterministic numerical comparison against the canonical estimate. The extraction judge sees the executed code, stdout, scenario context, and method family, but not the evaluated model identity, canonical estimate, or L2b+ label. A completed 50-cell blinded human audit of this extraction instrument finds 90.9% numeric agreement among human-scoreable effect-present cells and 88.6% agreement on the induced L2b+ pass/fail decision; Appendix E gives denominators and disagreement cases. The judge can still err, so we treat L2b+ as an audited execution-grounded metric rather than an infallible oracle.

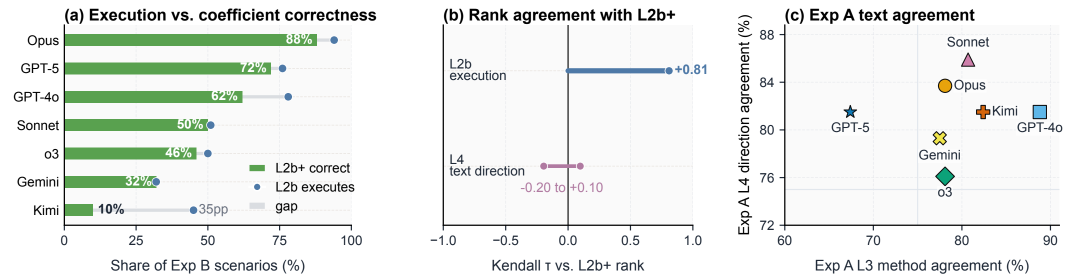


Figure 2: Headline result. **(a)** L2b execution and L2b+ coefficient correctness are distinct. **(b)** L2b ranking tracks L2b+ correctness (Kendall $\tau = +0.81$), while L4 agreement against consensus direction labels has weak association with Exp B L2b+ ranking ($\tau \in [-0.20, +0.10]$). **(c)** Exp A L3/L4 agreement across the seven primary models; L3/L4 are diagnostics, not executable correctness endpoints. Throughout the paper, model markers follow a fixed (color, shape) mapping: Opus ●, Sonnet ▲, GPT-4o ■, GPT-5 ★, o3 ⧫, Kimi $+$, Gemini $\times$.

**Threshold choice.** The default L2b+ tolerance is relative error at most 50%. This threshold is intentionally broad: it allows finite-sample and implementation variation while still rejecting estimates that are off by a factor large enough to change practical interpretation. The binary threshold is a coarse workflow-success endpoint; the relative-error profiles and tolerance sweep test whether conclusions depend on a single cutoff.

## 5 Results

The results follow the same separation as the benchmark design. Exp A first shows what real-paper text agreement can and cannot tell us. Exp B then tests the executable claim: whether the model-written workflow recovers the canonical estimate. Finally, calibration asks whether the models themselves can identify when that workflow is wrong under a retrospective self-report prompt.

**Finding 1: real-paper text agreement is useful but not decisive.** On Exp A, L3 is scoreable for 187/259 papers and L4 for 92/259 papers (unscoreable cases are predominantly papers where the 4-LLM consensus did not converge on a single canonical method-family label or positive/negative direction label). GPT-4o leads L3 (88.8%) and Sonnet leads L4 (85.9%); the rankings do not collapse to one textual ability score. The 30-paper blinded human ambiguity audit reinforces this caution: agreement with the 4-LLM consensus is only fair for direction labels (Cohen's $\kappa = 0.294$), so L4 scores partly reflect reference-label noise rather than purely model behavior. Figure 2(c) summarizes the L3/L4 separation; Figure 3 breaks L3 down by method family, and Event Study has the lowest observed cross-model L3 mean in this sample with smaller corpus size limiting precision. Exp A is therefore best read as a real-context text diagnostic, not as the paper's primary correctness endpoint.

**Finding 2: runnable code is not necessarily correct.** Figure 2 shows that L2b execution and L2b+ correctness separate the model panel. Final ES-aware canonical L2b+ rates span 10–88% across the seven primary commercial models: Opus 88%, GPT-5 72%, GPT-4o 62%, Sonnet 50%, o3 46%, Gemini 32%, and Kimi 10%. This spread is invisible if evaluation stops at whether the model wrote code. Appendix C.2 decomposes Exp B failures into execution, reporting, target-coefficient, event-window, IV-stage, and RDD cutoff/sign categories.

**Finding 3: execution ranking tracks reference-agreement correctness.** Across the seven models, the L2b ranking strongly matches the final L2b+ ranking (Kendall $\tau = +0.81$, scenario-clustered 95% bootstrap CI $[0.62, 0.90]$; Spearman $\rho = +0.93$, $[0.75, 0.96]$; 1,000 resamples, seed 20260507; every replicate exceeds the L4-vs-L2b+ upper bound $+0.10$). Appendix C.3 reports leave-one-model-out stability: Kendall $\tau$ remains in $[0.733, 0.867]$, with $\tau = 0.714$ when adding Llama as robustness-only. By contrast, L4 agreement against consensus direction labels has weak or negative association with Exp B L2b+ ranking ($\tau \in [-0.20, +0.10]$ across two frozen scorer variants). Execution is not sufficient, but it is a much stronger proxy for reference-agreement correctness than text-level direction. Appendix C reports $P(\text{L2b+} \mid \text{L2b})$ to separate execution failures from executed-but-wrong estimates.

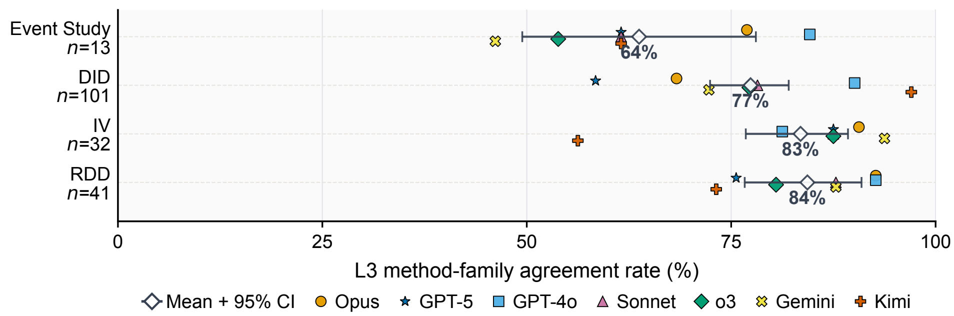


Figure 3: Exp A L3 method-family agreement by method family. Colored markers show the seven model-level agreement rates; black diamonds and whiskers show the cross-model mean with paper-cluster bootstrap 95% intervals. Event Study has the lowest observed mean (64%, $n = 13$); with $n = 13$ papers the cross-model 95% bootstrap CI is wide and Event Study agreement should be treated as exploratory rather than as a comparative finding. L3 is a text-agreement diagnostic, not an executable correctness endpoint.

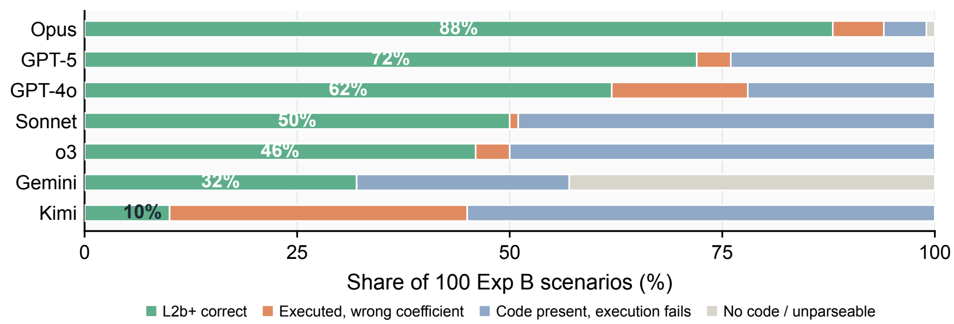


Figure 4: Execution cascade on Exp B. Each bar partitions 100 scenarios for one model into L2b+ correct estimates, executed-but-wrong coefficients, execution failures, and missing or unparseable code. The decomposition shows why runnable code alone is not sufficient.

**Finding 4: L2b+ rankings are not threshold artifacts.** The L2b+ relative-error distributions show that stronger models concentrate executed estimates near zero, while weaker models accumulate large-error tails. Thus the ranking is better understood as a distributional difference in numerical accuracy rather than an artifact of the default 50% cutoff. Appendix F reports the full relative-error profiles and tolerance sweep. Appendix C.4 reports 10%–100% tolerance sweeps; stricter cutoffs lower absolute pass rates but preserve the main text-versus-execution separation.

**Finding 5: naive retrospective confidence is a weak triage signal.** Calibration asks each model to rate its own previous Exp B output without seeing the L2b+ pass/fail label. Actual L2b+ rates vary sharply, but confidence gaps remain small for the main models. GPT-5 illustrates the asymmetry: it is second on L2b+ but has a slightly negative confidence gap. Under our retrospective confidence prompt, the self-assessment signal therefore does not provide a reliable deployment triage signal; this does not rule out stronger calibration interfaces. Appendix F reports reliability diagrams for the retrospective confidence protocol.

**Robustness check: the execution–correctness gap reproduces for an open-weights model.** As a single cross-vendor check, we evaluate Llama-3.3-70B-Instruct on the same 100 Exp B scenarios using the same prompt and scoring pipeline. It reaches L2b $= 41\%$ and final ES-aware L2b+ $= 20\%$, placing it between Kimi and Gemini and reproducing the same qualitative gap between runnable code and numerically correct code. Adding Llama as an eighth model lowers Kendall $\tau$ to $0.714$ (Appendix C.3) without changing the qualitative L2b–L2b+ ordering or L2b–L4 separation. We treat this as a robustness check, not a complete benchmark of the open-source ecosystem.

Table 3: Calibration summary against final L2b+ v2 correctness. Gap is mean numerical confidence on correct outputs minus mean confidence on incorrect outputs. Small gaps indicate weak triage value.

| Model | n | L2b+ v2 | Conf. gap |
|---|---|---|---|
| Opus | 100 | 88.0 | +0.075 |
| GPT-5 | 99 | 71.7 | -0.045 |
| GPT-4o | 100 | 62.0 | +0.016 |
| Sonnet | 99 | 49.5 | -0.011 |
| o3 | 100 | 46.0 | -0.010 |
| Gemini | 48 | 31.2 | +0.234 |
| Kimi | 100 | 10.0 | +0.027 |

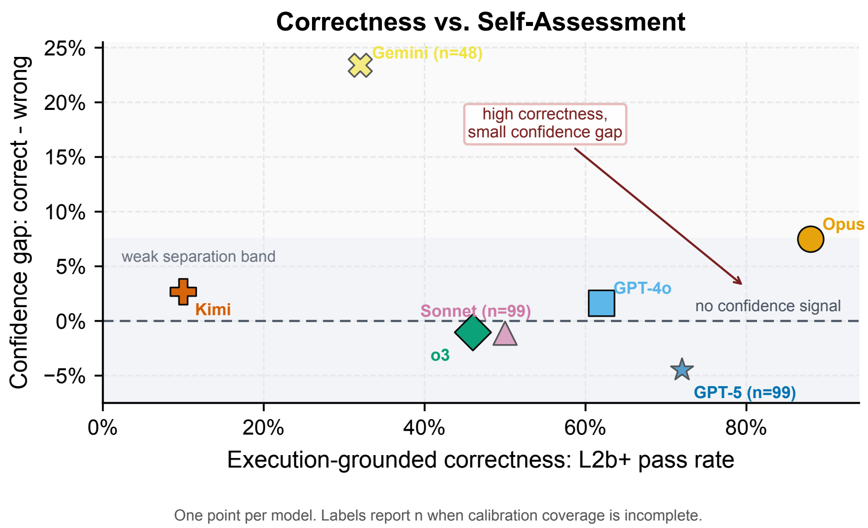


Figure 5: Correctness versus self-assessment. Each point is one model. The horizontal axis is execution-grounded L2b+ correctness; the vertical axis is the mean numerical-confidence gap between correct and incorrect workflows. High L2b+ accuracy does not imply a reliable confidence signal.

# 6 Reproducibility and Human Audits

The final submission build is tied to a frozen audit state. The released artifacts include reconstructed Exp A paper fields, four-LLM consensus reference labels, 1813 Exp A outputs, 700 primary Exp B model-scenario execution records, 100 realised CSV datasets with canonical-estimator outputs, 646 calibration records, prompt hashes, scorer decision records, and cached model responses. Deterministic scoring can be rerun without new LLM calls; because commercial APIs are not bit-reproducible, the release treats cached JSON outputs as the reproducible model-output layer. Appendix A lists artifact paths and checksums.

Human annotation is intentionally an ambiguity audit rather than a training or tuning input. A blinded 30-paper Exp A slice was labeled from the original PDFs after the automated benchmark freeze without access to Phase-2 LLM votes, consensus labels, or model outputs, and these human labels are not used to change frozen consensus labels, prompts, outputs, or scoring thresholds. Agreement with the four-LLM consensus is method 60.0% / direction 47.6% (Cohen's $\kappa = 0.606$ and $0.294$), supporting our interpretation of Exp A as a real-context agreement diagnostic with substantial ambiguity; Exp B supplies executable correctness.

A separate blinded human audit validates the coefficient-extraction instrument used for L2b+. The completed 50-cell audit finds 90.9% numeric agreement among human-scoreable effect-present cells and 88.6% agreement on the induced L2b+ pass/fail decision; Appendix E gives denominators and disagreement cases. This audit validates the measurement instrument but does not change the frozen headline L2b+ rates.

## 7 Limitations

First, Exp A labels are reference labels rather than L2b+ correctness labels: the final labels come from a four-LLM consensus, and the completed blinded 30-paper ambiguity audit agrees with consensus on 60.0% of method labels and 47.6% of direction labels, with Cohen's $\kappa = 0.294$ for direction. The audit is an ambiguity check, not a full human re-adjudication of the 259-paper corpus. The consensus pool also overlaps with the evaluated panel (two exact matches, two same-provider siblings; see ğ3), so L3/L4 are agreement diagnostics under structural overlap rather than independent text-correctness measurements; L2b+ in Exp B does not use this pool. Because Exp A uses published papers, we also cannot fully rule out model familiarity with some studies from pretraining or public metadata.

Second, Exp B is synthetic and structured. This is necessary for executable reference estimates, but it measures whether models can implement textbook workflows on known CSV data, not whether they can conduct an entire empirical research project. The DGPs cover four major method families but cannot represent all design choices in applied economics and finance. In particular, Exp B does not test literature review, data cleaning, design selection from ambiguous field constraints, or iterative robustness analysis.

Third, Exp B fixes R as the execution backend. L2b+ therefore measures executable workflow reliability under this released backend rather than language-invariant causal ability; backend sensitivity across Python, Stata, Julia, or other environments is an important extension.

Fourth, calibration is retrospective and prompt-specific. The protocol tests whether models can assess their own prior outputs under our confidence-elicitation prompt, not whether all possible uncertainty interfaces would fail. Calibration coverage is also incomplete for some models. The negative result should be read as evidence that naive retrospective self-reported confidence is not enough, not as proof that no calibrated interface can be designed.

Fifth, the evaluated model panel is limited. The primary panel consists of closed or hosted frontier systems, which is appropriate for the deployment question that motivates the paper but does not characterize the full open-source ecosystem. We add one open-weights model as a cross-vendor robustness check; a complete benchmark of multiple Llama, Qwen, Mistral, DeepSeek, and domain-specific systems is left for future work. Model-level rank correlations should also be read as descriptive diagnostics over this evaluated panel, not population-level estimates over all possible LLMs.

Sixth, L2b+ is an audited measurement instrument rather than an oracle. It depends on the canonical specification, the relative-error tolerance, event-study scale conversions, and coefficient extraction from executed stdout. We mitigate these risks through frozen scorer rules, decision records, and a completed coefficient-extraction human audit, but individual-cell mistakes remain possible and should be treated as part of the benchmark's measurement uncertainty.

Finally, the benchmark is intentionally single-shot and non-interactive. Models receive fixed prompts and data previews, and the benchmark scores the returned workflow rather than allowing extended human-in-the-loop debugging or iterative robustness analysis. This design makes evaluation reproducible and comparable across models, but it does not measure how a supervised analyst might use an LLM over multiple rounds to improve a causal study.

## 8 Conclusion

CAUSALVERIFY shows that the right unit of evaluation for LLM-assisted causal inference is the workflow: a model may produce plausible causal text and runnable code while still estimating the wrong causal quantity. Across seven evaluated LLMs, final L2b+ pass rates range from 10% to 88%; execution ranking tracks reference-agreement much better than L4 direction agreement, and naive retrospective confidence does not reliably separate correct from incorrect workflows. The question for causal inference is therefore not whether the model sounds right, but whether its workflow executes, agrees with a specified target, and carries audited uncertainty.

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# A Reproducibility Details

**Appendix roadmap.** Appendix A lists frozen artifacts and checksums. Appendix B describes the 30-paper Exp A ambiguity audit. Appendix C reports Exp B robustness diagnostics: canonical-estimator targets, conditional $P(\mathrm{L2b+} \mid \mathrm{L2b})$, rank-stability, tolerance sweep, scorer evolution, and method-family breakdown. Appendix D contains the descriptive RID pilot. Appendix E reports the 50-cell coefficient-extraction human audit. Appendix F contains additional diagnostic figures moved from the main text.

The final submission build is tied to a complete audit state. The audit tag `v11-freeze-2026-04-29` preserves the full provenance trail, while `causalverify_neurips2026.tex` provides a submission-length rendering of the same claims. The final submission PDF checksum is documented in `audit/SUBMISSION_BUILD_SUMMARY.md`.

Table 4: Frozen artifacts used by the final submission build. The full audit state remains available for inspection; the submission paper is a shorter rendering of the same frozen claims.

| **Claim** | **Frozen artifact** |
|---|---|
| Exp A outputs | `experiments/exp_a/outputs/` |
| Exp A scores | `experiments/exp_a/auto_scores.csv` |
| Exp B L2b+ | `l2b_plus_scores_canonical_judge_v2.csv` (primary 7; Llama R6 retained) |
| Calibration | `calibration_scores.csv`, `calibration_summary_v2.json` |
| Human validation | `audit/human_gold/human_vs_llm_consensus.md` |
| Paper checksum | `audit/SUBMISSION_BUILD_SUMMARY.md` |
| Gate evidence | `audit/V11_ACCEPTANCE_GATES.md` |

# B Human Annotation Details

The repository contains a paper-native labeling protocol and a completed blinded 30-paper ambiguity audit slice. The annotator labeled method family and direction from the original PDFs without access to Phase-2 LLM votes, consensus labels, or model outputs. One target paper (`paper_159`) was excluded because the available PDF was appendix-only; `paper_17` was added as a blinded replacement from the unsealed sample manifest, which contains only paper ID, difficulty tier, and PDF path. These human labels are not used to silently change the frozen Exp A labels. Human adjudication was conducted after the automated benchmark freeze as an ambiguity audit, not as a training, prompting, or tuning input; the labels are not used to construct prompts, select model outputs, or tune scoring thresholds.

# C Exp B Robustness Details

This appendix reports supporting diagnostics for Exp B. These tables are derived from `l2b_plus_scores_canonical_judge_v2.csv` and do not replace the frozen headline L2b+ rates in the main text.

## C.1 Canonical estimators for Exp B

L2b+ compares the model-reported estimate to a fixed realised-data reference estimate. The canonical estimator is a benchmark-defined reference path for executable evaluation, not a claim that only one empirical analysis is scientifically defensible. It is also distinct from the structural DGP parameter used to simulate the data.

Because L2b+ requires code execution, raw L2b–L2b+ association is partly structural. Table 6 therefore reports the conditional metric $P(\mathrm{L2b+} \mid \mathrm{L2b})$, which asks whether executed code computes the right coefficient.

Table 5: Benchmark-defined canonical estimators for Exp B L2b+ scoring.

| Method | Canonical reference estimate | Accepted model-reported coefficient | Common non-target coefficients | Scope limitation |
|---|---|---|---|---|
| DID | TWFE coefficient on `treat_x_post` after unit and period demeaning | `treat_x_post`, `treated:post`, or equivalent treatment-by-post effect | `treated` main effect, `post` main effect, placebo or pre-trend coefficient | Clean benchmark DID template, not a full benchmark of staggered or heterogeneous-treatment DID estimators |
| Event Study | Market-model coefficient on `event_day >= 0`; ES-aware v2 accepts frozen integer post-window conversions to this per-day scale | Explicit post-event AAR/CAR that can be deterministically converted to the canonical scale | Pre-event coefficient, arbitrary non-frozen window, diagnostic or placebo return | Fixed benchmark window mapping, not a universal event-study estimand |
| IV | Realised-data 2SLS coefficient on `x`, instrumented by `z`, with controls exogenous when present | Second-stage coefficient on `x` or the named endogenous variable | First-stage coefficient on `z`, reduced form, or control coefficient | Tests implementation of the IV estimand; does not validate exclusion restriction or weak-instrument diagnostics |
| RDD | Local-linear cutoff effect on the realised data: coefficient on `treated` near cutoff 0.5 with running-variable slope and interaction | Treatment, above-cutoff, or discontinuity coefficient consistent with benchmark direction | Density test, bandwidth statistic, unrelated global polynomial trend, placebo cutoff estimate | Fixed benchmark RDD reference, not an exhaustive comparison of bandwidth or inference choices |

Table 6: Conditional Exp B correctness among executed outputs. "Wrong after execution" is L2b count minus L2b+ v2 count.

| Model | L2b | L2b+ | Wrong after exec. | $P$(L2b+\|L2b) | Gap |
|---|---|---|---|---|---|
| Opus | 94 | 88 | 6 | 93.6% | 6 pp |
| GPT-5 | 76 | 72 | 4 | 94.7% | 4 pp |
| GPT-4o | 78 | 62 | 16 | 79.5% | 16 pp |
| Sonnet | 51 | 50 | 1 | 98.0% | 1 pp |
| o3 | 50 | 46 | 4 | 92.0% | 4 pp |
| Gemini | 32 | 32 | 0 | 100.0% | 0 pp |
| Kimi | 45 | 10 | 35 | 22.2% | 35 pp |

## C.2 Exp B failure-taxonomy diagnostic

We also audit the non-L2b+ mass with a conservative rule-based taxonomy. The taxonomy is diagnostic rather than a scoring replacement: it distinguishes missing or unparseable code, execution failures, executed-but-unreported effects, wrong target coefficients, event-window or scale mismatches, IV stage errors, and RDD cutoff or sign-convention issues. Ambiguous cases are left uncategorized rather than forced into a narrow label.

Table 7: Exp B failure-taxonomy diagnostic, per primary model. Cells show counts and (share of that model's non-L2b+ failures). Per-model breakdown reveals that strong models execute correctly when they execute (Opus, Sonnet, Gemini conditional pass rates $\geq 93\%$; see Table 6), while weaker models more often produce executed-but-wrong code (Kimi 22.2%, GPT-4o 79.5% conditional pass). Per-model rows sum to 340 non-L2b+ cells, matching the aggregate decomposition (Exec fail 230, No code 44, Wrong coef 26, Coef-not-reported 15, Window/scale 11, IV stage 5, RDD 3, Other 6).

| Model | n | Exec fail | No code | Wrong coef | Coef N/R | Win/scale | IV stage | RDD | Other |
|---|---|---|---|---|---|---|---|---|---|
| Opus | 12 | 5 (41.7%) | 1 (8.3%) | 2 (16.7%) | 0 | 3 (25.0%) | 0 | 0 | 1 (8.3%) |
| Sonnet | 50 | 49 (98.0%) | 0 | 1 (2.0%) | 0 | 0 | 0 | 0 | 0 |
| GPT-4o | 38 | 22 (57.9%) | 0 | 3 (7.9%) | 1 (2.6%) | 3 (7.9%) | 4 (10.5%) | 0 | 5 (13.2%) |
| o3 | 54 | 50 (92.6%) | 0 | 2 (3.7%) | 1 (1.9%) | 1 (1.9%) | 0 | 0 | 0 |
| Kimi | 90 | 55 (61.1%) | 0 | 18 (20.0%) | 13 (14.4%) | 3 (3.3%) | 1 (1.1%) | 0 | 0 |
| Gemini | 68 | 25 (36.8%) | 43 (63.2%) | 0 | 0 | 0 | 0 | 0 | 0 |
| GPT-5 | 28 | 24 (85.7%) | 0 | 0 | 0 | 1 (3.6%) | 0 | 3 (10.7%) | 0 |

## C.3 Rank-stability diagnostic

The rank correlations in Figure 2 are descriptive model-level diagnostics over seven primary models rather than population-level estimates. To assess panel sensitivity, we recompute Kendall $\tau$ and Spearman $\rho$ after leaving out each model in turn. The leave-one-model-out Kendall range is $0.733$–$0.867$ and the Spearman range is $0.886$–$0.943$. Adding Llama-3.3-70B-Instruct as a robustness-only eighth model gives Kendall $\tau = 0.714$ and Spearman $\rho = 0.881$, while the primary leaderboard remains the seven-model panel.

Table 8: Rank-stability diagnostic for the Exp B model-level correlations. Primary-panel rows summarize the audit table in `paper/tables/exp_b_rank_stability_primary7.csv`; the Llama row is robustness-only and does not enter the primary leaderboard.

| Diagnostic | $N$ models | Kendall $\tau$ | Spearman $\rho$ |
|---|---|---|---|
| Primary seven-model baseline | 7 | 0.810 | 0.929 |
| Leave-one-model-out range | 6 each | 0.733–0.867 | 0.886–0.943 |
| Primary seven plus Llama robustness-only | 8 | 0.714 | 0.881 |

Table 9: Scorer evolution on Exp B. Regex extraction is retained as a legacy diagnostic; the final headline uses the ES-aware v2 score. Entries are pass rates in percentage points.

| Model | Regex | Judge | ES-aware v2 | v2–Judge |
|---|---|---|---|---|
| Opus | 64 | 71 | 88 | +17 |
| GPT-5 | 29 | 59 | 72 | +13 |
| GPT-4o | 28 | 50 | 62 | +12 |
| Sonnet | 27 | 31 | 50 | +19 |
| o3 | 18 | 37 | 46 | +9 |
| Gemini | 15 | 29 | 32 | +3 |
| Kimi | 6 | 8 | 10 | +2 |

Table 10: L2b+ v2 by model and method family. Each cell reports passes/available scenarios.

| Model | DID | ES | IV | RDD |
|---|---|---|---|---|
| Opus | 29/30=97% | 19/24=79% | 23/24=96% | 17/22=77% |
| GPT-5 | 20/30=67% | 17/24=71% | 23/24=96% | 12/22=55% |
| GPT-4o | 25/30=83% | 13/24=54% | 15/24=62% | 9/22=41% |
| Sonnet | 20/30=67% | 21/24=88% | 3/24=12% | 6/22=27% |
| o3 | 8/30=27% | 12/24=50% | 16/24=67% | 10/22=45% |
| Gemini | 14/30=47% | 5/24=21% | 13/24=54% | 0/22=0% |
| Kimi | 5/30=17% | 4/24=17% | 1/24=4% | 0/22=0% |

## C.4 Tolerance-sweep diagnostic

The binary L2b+ endpoint uses a broad default relative-error tolerance of 50%. To check whether the qualitative conclusion depends on this cutoff, we recompute primary-seven pass rates at 10%, 25%, 50%, 75%, and 100%. The sweep is used as a robustness diagnostic; the frozen headline results continue to use the default 50% tolerance. Table 11 reports all-scenario pass rates, so non-executing outputs remain failures.

Table 11: Tolerance-sweep diagnostic for Exp B L2b+ scoring. Each cell reports the pass rate out of 100 scenarios; parentheses give the model rank at that tolerance. The 50% column is the frozen headline threshold.

| **Model** | **10%** | **25%** | **50%** | **75%** | **100%** |
|---|---|---|---|---|---|
| Opus | 72% (1) | 84% (1) | 88% (1) | 88% (1) | 88% (1) |
| GPT-5 | 64% (2) | 71% (2) | 72% (2) | 72% (2) | 72% (2) |
| GPT-4o | 53% (3) | 59% (3) | 62% (3) | 63% (3) | 63% (3) |
| Sonnet | 42% (4) | 48% (4) | 50% (4) | 50% (4) | 50% (4) |
| o3 | 38% (5) | 44% (5) | 46% (5) | 47% (5) | 47% (5) |
| Gemini | 32% (6) | 32% (6) | 32% (6) | 32% (6) | 32% (6) |
| Kimi | 9% (7) | 9% (7) | 10% (7) | 13% (7) | 14% (7) |

# D RID Pilot Ablation

Experiment C is a small descriptive pilot that reruns a 10-paper subset under the RID pre-commitment protocol. It is not used for the headline benchmark claims, because $N = 10$ per model is too small for model-level inference. We include it only as an audit artifact showing how pre-commitment can change text-level L3/L4 outcomes in either direction.

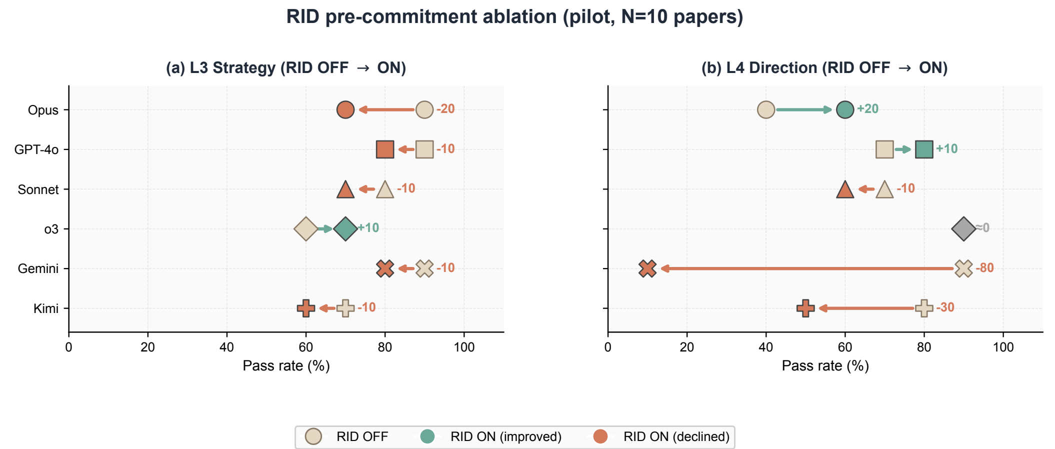


Figure 6: RID pre-commitment ablation pilot ($N = 10$ papers per model, descriptive only). Dots compare RID OFF and RID ON pass rates for L3 strategy and L4 direction. No individual comparison is statistically significant; the pilot is retained as an audit artifact rather than as evidence for the main claims.

# E Coefficient-Extraction Human Audit

The coefficient-extraction judge is a measurement instrument: it identifies the scalar treatment-effect coefficient from executed R code/stdout before L2b+ compares that coefficient to the canonical estimator. The repository contains a blinded human-validation audit in `audit/l2b_judge_human_validation/`. The annotation form hides model identity, L2b+ label, canonical estimate, judge effect, regex effect, and ranking information. The completed audit covers 50 primary-panel L2b=1 cells sampled with seed 20260502: DID 13, Event Study 12, IV 11, and RDD 14. Among comparable audited cells, it found 90.9% numeric agreement and 88.6% agreement on the induced L2b+ pass/fail decision. Disagreements concentrate in event-study window choices and RDD sign/printing ambiguities. This audit supports the coefficient-extraction instrument; it does not change the frozen headline L2b+ rates.

**Disagreement categories.** Most disagreements arise from two sources: (i) Event Study outputs that report a cumulative post-event window rather than the canonical per-day scale, and (ii) RDD outputs where printed signs, jump orientation, or robustness estimates make the main effect ambiguous. These cases motivate treating L2b+ as an audited measurement instrument rather than an infallible oracle.

# F Additional Diagnostics

This appendix collects supporting diagnostic figures for the main results. The main text reports the headline conclusions; the relative-error profiles, calibration reliability diagrams, and L4 scorer-operationalization diagnostic below support those conclusions but are not required for the central correctness argument.

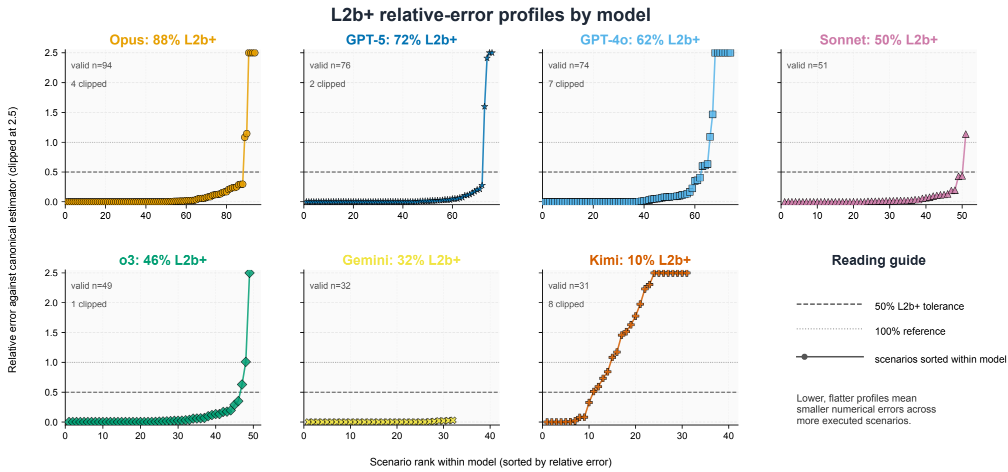


Figure 7: Additional Exp B robustness diagnostic. Each panel sorts executed estimates by relative error against the realised-data canonical estimator; the dashed line marks the default 50% tolerance. These profiles support the main-text claim that L2b+ rankings are not driven by a single cutoff.

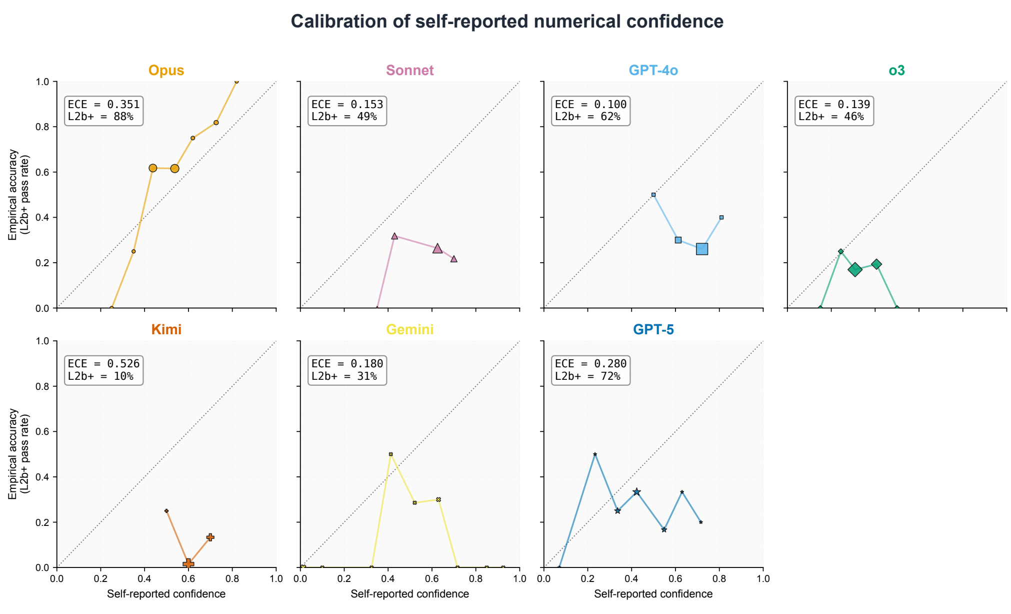


Figure 8: Additional calibration diagnostic. Reliability diagrams compare retrospective self-reported confidence with empirical L2b+ correctness. These diagnostics support the main-text finding that naive retrospective self-reported confidence does not reliably separate correct from incorrect workflows under our retrospective confidence prompt. 95% bootstrap CI on ECE (1,000 resamples within model): see `paper/derived_analyses/ece_bootstrap_ci.csv`.

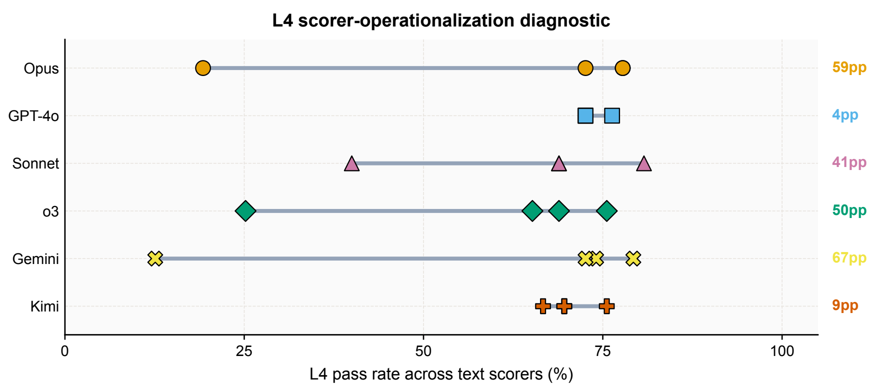


Figure 9: L4 scorer-operationalization diagnostic. Alternative frozen L4 text-direction scorers produce unstable pass rates across models. This diagnostic is not part of the primary seven-model headline figure because the scorer sweep was run on the six models with frozen scorer-variant outputs.

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction define the scope as a benchmark for verifying LLM causal-inference workflows. They distinguish real-paper text agreement from execution-grounded coefficient correctness, and state that Exp A measures agreement diagnostics while Exp B provides the executable correctness endpoint.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Limitations section covers Exp A label ambiguity, the 30-paper human ambiguity audit, the fact that Exp A is not a fully human-adjudicated reference-label corpus, the synthetic scope of Exp B, the dependence of L2b+ on canonical estimators and coefficient extraction, model snapshot limits, and domain scope.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

   Answer: [N/A]

   Justification: This is an empirical benchmark paper; no theoretical results or proofs are presented.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: [Yes]

   Justification: The anonymized release includes 259 active Exp A paper records, four-LLM consensus labels, 100 Exp B synthetic DGP scenarios with realised CSV data files and canonical-estimator targets, 1813 Exp A outputs, 700 Exp B primary execution cells, 646 calibration records, the L2b+ coefficient-extraction pipeline, prompt templates, scoring scripts, and frozen audit summaries.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient documentation and instructions for reproducing the main experimental results?

   Answer: [Yes]

   Justification: The anonymized repository submitted with the paper contains the code, frozen outputs, documentation, datasheet, component-level data-license notes, and reproduction commands. The README explains how to rerun deterministic scoring for Exp A, Exp B, and calibration without making new LLM API calls.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: [Yes]

   Justification: No training is performed; this is an evaluation-only benchmark. The paper and release specify the evaluated models, prompt construction, Exp A consensus-label construction, Exp B DGP generation, realised-data canonical estimators, L2b/L2b+ scoring, and calibration protocol. Prompt templates and implementation details are included in the code release.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The paper reports sample sizes for all scoreable subsets, model-level pass rates, rank correlations for the L2b/L2b+ and L4/L2b+ comparisons, and a relative-error distribution analysis showing that the main Exp B ranking is not an artifact of a single tolerance threshold. The paper does not make pairwise model-significance claims.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type, memory, time, etc.) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The original model-generation runs use commercial LLM API calls and no local GPU compute. The deterministic scoring and figure-generation steps can be rerun from frozen outputs on a standard CPU environment. Token counts, costs, prompts, and run logs are preserved in the audit artifacts.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: The paper evaluates publicly available LLMs on published economics papers and synthetic scenarios. No human subjects are involved. The broader impact paragraph (Section 8) discusses risks of using LLMs for policy-facing causal analysis.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: [Yes]

    Justification: Section 8 (Broader impact paragraph) discusses the risk that LLMs used in policy-facing causal analysis without recognizing documented failure modes could support misleading conclusions. The benchmark itself is designed to surface these failure modes, which is a positive impact.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

    Answer: [N/A]

    Justification: The released data consists of published paper metadata and synthetic scenarios; the released code is an evaluation framework. Neither poses a meaningful misuse risk.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models) used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The 259 active paper records include source metadata and provenance. Models are identified by provider and API version in the release documentation. The code is released under the repository's code license, synthetic DGP artifacts and scored outputs are licensed as documented in the data-license file, and original published article PDFs are not relicensed by this repository unless redistribution rights are explicitly confirmed. OpenAlex-derived metadata is used under its stated terms.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets in a format that is easily accessible?

Answer: [Yes]

Justification: The benchmark is accompanied by a DATASHEET.md following Gebru et al. (2021), a detailed README with reproduction instructions, and structured JSON schemas for all papers and scenarios.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots?

    Answer: [N/A]

    Justification: No crowdsourcing or human subjects research was conducted. A 30-paper blinded author ambiguity audit and a 50-cell blinded coefficient-extraction audit were performed after the automated benchmark freeze; neither was used for training, prompt tuning, model selection, or threshold tuning. The labeling protocols and instructions are released in the repository.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether compensation was provided, etc.?

    Answer: [N/A]

    Justification: No human subjects research was conducted.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs in the research process?

    Answer: [Yes]

    Justification: LLMs are the subject of evaluation and are used in the benchmark pipeline for Exp A consensus labeling, field reconstruction, coefficient extraction, and calibration elicitation. The release includes prompts, model outputs, caches, and audit records needed to inspect those uses. The authors also used LLM assistance during coding and manuscript editing, with final responsibility for claims and text retained by the authors.